

Computational Mechanics

2. Predicting the future

Exercise sheet

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Instructions

Give a clear justification for each answer. You may use results proved in the lecture unless an exercise asks you to establish them. Questions marked **Extension** are optional.

Exercise 1

From statistics to partitions

Let $u_1, \dots, u_6 \in \mathcal{X}_+^*$ be six possible histories. Suppose that they induce the following conditional distributions over the entire future:

history	u_1	u_2	u_3	u_4	u_5	u_6
future distribution	P	P	P	Q	Q	S

where P , Q , and S are pairwise distinct.

Three statistics R_A , R_B , and R_C induce the partitions

$$\begin{aligned}\Pi_A &= \{\{u_1, u_2\}, \{u_3\}, \{u_4\}, \{u_5\}, \{u_6\}\}, \\ \Pi_B &= \{\{u_1, u_2, u_3\}, \{u_4, u_5\}, \{u_6\}\}, \\ \Pi_C &= \{\{u_1, u_2, u_3, u_4\}, \{u_5\}, \{u_6\}\}.\end{aligned}$$

Recall that a statistic is sufficient for prediction when histories assigned the same value induce the same distribution over the future.

- What does it mean for two histories to belong to the same cell of the partition induced by a statistic? Illustrate your answer using u_1 and u_2 in Π_A .
- Determine which of R_A , R_B , and R_C are sufficient. If a statistic is not sufficient, give two histories witnessing the failure.
- Which cells of Π_A can be merged without losing sufficiency? Perform all such merges and identify the resulting partition.
- Explain why no two cells of Π_B can be merged while preserving sufficiency. What does this show about R_B ?
- Prove that every refinement of a sufficient partition is also sufficient. Is every coarsening of a sufficient partition necessarily sufficient? Use Π_A , Π_B , and Π_C to illustrate your answer.

(f) **Extension.** Consider the partition

$$\Pi_D = \{\{u_1\}, \{u_2, u_3\}, \{u_4, u_5\}, \{u_6\}\}.$$

Show that Π_A and Π_D are both sufficient but that neither refines the other. How can both nevertheless refine the same minimal sufficient partition?

Exercise 2

Causal states are minimal sufficient statistics

For $x, x' \in \mathcal{X}_+^*$, define the relation $\sim$ by

$$x \sim x' \iff \Pr(X_{|x|+1:|x|+|y|} = y \mid X_{1:|x|} = x) = \Pr(X_{|x'|+1:|x'|+|y|} = y \mid X_{1:|x'|} = x')$$

for every $y \in \mathcal{X}^*$. Define the causal state of x by

$$\epsilon(x) := [x]_{\sim},$$

and write $\mathcal{S} := \epsilon(\mathcal{X}_+^*)$ for the set of causal states.

(a) Prove that $\sim$ is an equivalence relation on $\mathcal{X}_+^*$ by establishing each of the following:

- (i) reflexivity: $x \sim x$ for every $x \in \mathcal{X}_+^*$;
- (ii) symmetry: if $x \sim x'$, then $x' \sim x$;
- (iii) transitivity: if $x \sim x'$ and $x' \sim x''$, then $x \sim x''$.

(b) Prove that ϵ is sufficient for prediction.

(c) Let $R : \mathcal{X}_+^* \rightarrow \mathcal{R}$ be any sufficient statistic. Prove that

$$R(x) = R(x') \implies \epsilon(x) = \epsilon(x').$$

Deduce that every sufficient partition refines the causal-state partition.

(d) Using parts (b) and (c), conclude that ϵ is a minimal sufficient statistic.

(e) **Extension.** Let T be another minimal sufficient statistic. Prove that T and ϵ induce the same partition of $\mathcal{X}_+^*$. Deduce that there is a bijection

$$\psi : \mathcal{S} \rightarrow \text{im}(T)$$

satisfying $T = \psi \circ \epsilon$.

Exercise 3

From finite generators to causal states

In this exercise, S_t denotes the hidden state immediately before the transition that emits X_{t+1} . All parameters lie strictly between 0 and 1.

(a) **The Zero–One–Random Process.** Let S_0 be uniformly distributed over $\{s_0, s_1, s_R\}$. The process has transitions

$$s_0 \xrightarrow{0:1} s_1, \quad s_1 \xrightarrow{1:1} s_R, \quad s_R \begin{cases} \xrightarrow{0:1-p} s_0, \\ \xrightarrow{1:p} s_0. \end{cases}$$

An edge label $x : q$ means that the process emits x and follows the edge with probability q .

- (i) List all possible histories of length two. For each history, determine whether it uniquely identifies the current hidden phase. When it does, identify that phase.
- (ii) For each ambiguous length-two history, determine which additional observations resolve the ambiguity. Identify the resulting phase and explain why it remains known after every subsequent observation.
- (iii) Assume that $\epsilon(\emptyset)$, $\epsilon(0)$, $\epsilon(1)$, and $\epsilon(10)$ are pairwise distinct. Show that the causal states have shortest-length representatives

$$\emptyset, \quad 0, \quad 1, \quad 10, \quad 00, \quad 01, \quad 11.$$

Explain why no additional causal states arise.

- (iv) Under the assumption in part (iii), draw the automaton associated with the causal-state dynamics.

(b) **The Even Process.** Let S_0 be uniformly distributed over $\{A, B\}$ and consider the generator

$$A \xrightarrow{0:q} A, \quad A \xrightarrow{1:1-q} B, \quad B \xrightarrow{1:1} A.$$

Thus every run of 1s between successive 0s has even length.

- (i) For each of the length-one histories 0 and 1, determine whether the current hidden state can be identified exactly.
- (ii) Starting from each ambiguous history in part (i), consider its possible one-symbol extensions. Which extensions resolve the ambiguity, and which remain ambiguous? You may use the fact that $\epsilon(\emptyset) = \epsilon(11)$. What does this imply for histories of the form 1^n ?
- (iii) Assume additionally that $\epsilon(\emptyset) \neq \epsilon(1)$. Show that the causal states have shortest representatives

$$\emptyset, \quad 1, \quad 0, \quad 01,$$

and explain why there are no additional causal states. You may distinguish causal states using possible and impossible future continuations.

- (iv) Using $\epsilon(\emptyset) = \epsilon(11)$ and assuming $\epsilon(\emptyset) \neq \epsilon(1)$, draw the automaton associated with the causal-state dynamics.

(c) **The Simple Nonunifilar Source.** Let S_0 be uniformly distributed over $\{A, B\}$ and consider

$$A \xrightarrow{0:1/2} A, \quad A \xrightarrow{0:1/2} B, \quad B \xrightarrow{0:1/2} B, \quad B \xrightarrow{1:1/2} A.$$

- (i) Show that observing a 1 identifies the new hidden state as A .
- (ii) Starting from A , show that there are $n+1$ hidden paths that emit 0^n , each with probability 2^{-n} . How many of these paths can subsequently emit 1? Hence calculate

$$\Pr(X_{n+2} = 1 \mid X_{1:n+1} = 10^n).$$

- (iii) Show that this probability is different for every $n \geq 0$. Conclude that the histories

$$1, 10, 100, \dots$$

belong to distinct causal states and that the process has infinitely many causal states.

Exercise 4

Why next-token prediction is enough

Recall that

$$\mathcal{X}_+^* := \{x \in \mathcal{X}^* : \Pr(X_{1:|x|} = x) > 0\}$$

is the set of allowable histories.

Consider a model with parameters θ satisfying

$$h_n = f_\theta(h_{n-1}, x_n), \quad y_n = g_\theta(h_{n-1}, x_n).$$

Fix the initial hidden state h_0 and suppose that the model predicts the next-token distribution exactly: for every $n \geq 1$,

$$y_n = \Pr(X_{n+1} \mid X_{1:n} = x_{1:n}), \quad \forall x_{1:n} \in \mathcal{X}_+^*.$$

For every nonempty allowable history, define

$$R(x_{1:n}) := (h_{n-1}, x_n).$$

For $a \in \mathcal{X}$, define the deterministic update

$$\Phi_a(h, x) := (f_\theta(h, x), a).$$

For a word $z_{1:j}$, write

$$\Phi_{z_{1:j}} := \Phi_{z_j} \circ \dots \circ \Phi_{z_1}, \quad \Phi_{z_{1:0}} := \text{id}.$$

(a) Show that, whenever $x_{1:n}z_{1:j}$ is allowable,

$$R(x_{1:n}z_{1:j}) = \Phi_{z_{1:j}}(R(x_{1:n})).$$

Hence show that, if $R(x_{1:n}) = R(x'_{1:m})$ and both extended histories are allowable, then

$$R(x_{1:n}z_{1:j}) = R(x'_{1:m}z_{1:j}).$$

(b) Let $R(x_{1:n}) = R(x'_{1:m})$ and fix $z_{1:k} \in \mathcal{X}^k$. As the symbols of $z_{1:k}$ are read from left to right, use part (a) and exact next-token prediction to show that for each $0 \leq j < k$ such that the prefixes through $z_{1:j}$ are allowable,

$$\begin{aligned} \Pr(X_{n+j+1} = z_{j+1} \mid X_{1:n+j} = x_{1:n}z_{1:j}) \\ = \Pr(X_{m+j+1} = z_{j+1} \mid X_{1:m+j} = x'_{1:m}z_{1:j}). \end{aligned}$$

Deduce that either the same first zero-probability extension is encountered after both histories, or every prefix of $z_{1:k}$ is allowable after both histories.

(c) In the case where every prefix is allowable, use the conditional chain rule to write

$$\Pr(X_{n+1:n+k} = z_{1:k} \mid X_{1:n} = x_{1:n}) = \prod_{j=0}^{k-1} \Pr(X_{n+j+1} = z_{j+1} \mid X_{1:n+j} = x_{1:n}z_{1:j}).$$

Write the analogous product after $x'_{1:m}$ and compare its factors. Deal separately with the zero-probability case from part (b), and conclude directly that

$$R(x_{1:n}) = R(x'_{1:m}) \implies$$

$$\Pr(X_{n+1:n+k} = z_{1:k} \mid X_{1:n} = x_{1:n}) = \Pr(X_{m+1:m+k} = z_{1:k} \mid X_{1:m} = x'_{1:m}).$$

Hence $R(x_{1:n}) = (h_{n-1}, x_n)$ is a sufficient statistic for predicting the entire future.